

Muhammad Sharif

Full Stack Application Developer | Frontend Architecture | Enterprise SaaS | Payments & Compliance | Mobile
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SUMMARY

- Full stack application developer with production experience across enterprise compliance SaaS (Horuz), payments and checkout platforms (Paybilt), cross-platform mobile (Oasis), and full-stack web products — specializing in **React**, **TypeScript**, **Python**, **FastAPI**, **MongoDB**, **WebSockets**, **Docker**, and **CI/CD**.
- **Led** frontend architecture for a multi-tenant compliance platform, **modernizing** six core V1 modules into V2 while migrating Create **React** App to **Vite** — **reducing** total bundle size from **~6.6MB** to **~3.9MB**, initial **JavaScript** from **~3MB** to **~998KB**, and **improving** load times up to **~75%**.
- **Engineered** page-segmented document virtualization with **TanStack** Virtual and paginated APIs, enabling **200K+** records per **~20Mpx** scroll segment and stress-testing near **1M** rows without UI freezing — combining deduplication, cache-aware segments, and dynamic row sizing to work around browser scroll-height limits.
- **Delivered** high-stakes payments work at Paybilt: **built** a Vanilla JS SDK in **48 hours** (5,000+ lines) that **saved** a **\$1M+** Allianz partnership, **earned** 4 promotions in 12 months, and **shipped** compliance, checkout, and template systems supporting **\$1M+** in weekly transactions across government and enterprise clients.
- Self-driven engineer with end-to-end ownership from architecture through deployment — **refactoring** auth to **Zustand**, **designing** reusable component libraries (**Ant Design**, **Radix UI**, WCAG), and collaborating with product, design, backend, QA, and leadership in Agile/SCRUM teams.

EXPERIENCE

149MKM Software | Sept. 2025 – Present

Full Stack Application Developer — Lead Frontend, Horuz | Remote / Calgary, AB

Enterprise compliance SaaS (Horuz) for engineering, legal, construction, and audit workflows — obligations, evidence, approvals, digital signing, and multi-tenant RBAC.

- **Led** frontend architecture for Horuz, **modernizing** six core V1 modules into V2 (Dashboard, Obligations, Frameworks, Authentication, Actions, Files) to improve scalability, maintainability, responsiveness, and enterprise UX consistency across multi-tenant environments.
- **Migrated** the application from Create **React** App to **Vite** with lazy loading, route-level code splitting, viewport-based virtualization, and caching — **reducing** total bundle size from **~6.6MB** to **~3.9MB**, initial **JavaScript** from **~3MB** to **~998KB**, and **improving** load times up to **~75%**.
- **Engineered** a custom page-segmented virtualization system with **React** and **TanStack** Virtual, combining paginated API loading, virtual row rendering, deduplication, cache-aware segments, and dynamic pixel/row sizing — supporting **200K+** records per **~20Mpx** scroll segment and stress-testing near **1M** rows without UI freezing, delayed selection, or data loss.
- **Refactored** authentication and global state with **Zustand** stores, structured auth bootstrapping, session management, and role-aware access patterns — **reducing React** Context dependency and strengthening **JWT**, **RBAC**, and future **MFA** workflows.
- **Delivered** enterprise dashboards and the Files module with viewport-based document virtualization, lazy loading for **AWS**-hosted documents, client-side Excel exports, and secure downloads — **reducing** redundant API calls by **~50%**.
- Contributed to Rights Administration (RA), a cross-service admin platform for permissions across Horuz services using **JWT**, **RBAC**, **React**, **TypeScript**, **FastAPI**, **MongoDB**, **Docker**, and **CI/CD** — **cutting** deployment time by **~40%** and strengthening security compliance across role-based access workflows.
- **Built** performant, accessible UI systems and reusable component libraries with **React**, **Ant Design**, **Radix UI**, **Chart.js**, **Recharts**, and WCAG best practices for compliance tracking and operational dashboards.
- Connected frontend layers to **FastAPI** backends, **MongoDB**, **AWS**-hosted file infrastructure, WebSocket monitoring streams, and **Docker**-based **CI/CD** — collaborating with the CTO on architecture from design through production deployment.
- **Integrated** WebSocket-driven real-time monitoring for live logs, document status updates, and operational dashboards — pairing **React** hooks with async APIs for reactive compliance and audit workflows.

Reference: Available upon request

Paybilt | Apr. 2024 – Sept. 2025

Full Stack Developer | Hamilton, ON

Payment technology: checkout, merchant SDKs, compliance apps, and transaction systems for enterprise merchants, iGamingOntario operators, political fundraising, and insurance partners.

- **Built** a custom Vanilla JS payments SDK in **48 hours** (5,000+ lines in one weekend) with CSS theming, dynamic prefill, locked fields, and API-triggered payments — **saving** a **\$1M+** Allianz Insurance deal and **reducing** merchant onboarding time by **~70%**.
- **Earned** 4 promotions in 12 months (Junior Frontend Intern → Full Stack Developer) by consistently **delivering** production payments, compliance, and checkout systems under tight business timelines.

- Served as frontend lead on OCC V1, the Conservative Party of Canada donation platform **processing \$500K+** in contributions — **building** reusable **TypeScript/React/Next.js** components with performance and accessibility across devices.
- Managed and **led a 4-person** frontend team to deliver OCC V2, a one-click checkout experience that **increased** conversion rates by **25%**.
- **Built a Jinja2 + Flask** compliance app for iGamingOntario with a **Python** backend endpoint for payout payloads — enforcing legal name-matching, intercepting mismatched transactions, and **reducing** compliance risk for merchants including Bet365 and BetMGM (thousands of secure payouts weekly).
- **Designed** a unified Transaction Tracking interface rendering **300+** payment-status combinations from one adaptive **Jinja2** template — **cutting** frontend duplication by **~90%** and **reducing** support tickets related to unclear transaction status by **~40%**.
- **Engineered** the Template Registry Service (**FastAPI, MongoDB, Next.js, GraphQL, TypeScript**) powering **100+** versioned **Jinja2** templates through a flexible API, with **Flask** endpoints supporting dynamic success-screen rendering.
- Contributed to 5 live production systems across government and enterprise accounts, directly supporting **\$1M+** in weekly transactions.
Reference: Kevin Barthelot, Lead PM / Product Owner — 437-667-4562

Oasis | 2025

Mobile Application Developer (Contract) | Remote (KSA – Saudi Arabia)

Cross-platform Flutter real estate app aggregating listings, brokers, and clients.

- **Co-designed and built** a full-stack **Flutter** real estate application from scratch using Clean Architecture and **BLoC** — **delivering** 22+ screens across iOS, Android, and Web with property search, maps, favorites, comparisons, and price history.
- Implemented real-time broker chat with **WebSockets** (typing indicators, read receipts, online/offline status, automatic reconnection) **integrated** with **BLoC** state management.
- **Reduced** API calls by **~60%** through intelligent caching and request deduplication; **reduced** image payload sizes by **~70%** via compression and optimized upload flows.
- Collaborated with a backend team on secure **REST** integration for authentication, user roles, chat logs, and AI-powered chat features — **delivered** as a paid contract meeting production requirements.

Lotus | Jan. 2024

Frontend Developer | Toronto, ON

Web3-enabled startup landing experience with AWS-backed user preferences.

- **Crafted** a Web3-enabled landing experience with **Next.js, React**, and **Tailwind CSS**, **integrated** with an **AWS**-backed store for user preferences and NFT-driven engagement features.
- **Delivered** secure management of user preference and music-choice data within the application UI and integration layer.

Zam Zam Bags and Boutique | Feb. 2023 – May 2023

Full Stack Developer | Toronto, ON

Small-business e-commerce and catalogue site (zamzambandb.com).

- **Designed** and developed a responsive e-commerce website with **JavaScript, React.js**, Firebase/Firestore, HTML, CSS, and Bootstrap — **improving** user experience and business operations for a local retailer.
- **Built** administrative CRUD tooling so the owner could manage the product catalogue, add products, and edit listings independently without developer support.
- Conducted manual unit testing and debugging while optimizing API calls and database operations; managed the full Agile lifecycle from concept through deployment under tight deadlines.

MMM Diagnostics | Jan. 2022 – Sept. 2023

Data Entry Supervisor | Chicago, IL

Telehealth and diagnostics operations with HIPAA-sensitive data workflows.

- Contributed to a telehealth software initiative by translating business requirements into functional specifications and mapping workflows for clinical stakeholders.
- **Led** night-shift operations for a diagnostics team, verifying and entering an average of 80 tests per day under HIPAA-sensitive protocols with consistent accuracy.

Reference: Dr. Atif Rashid — +1 (616) 227-1404

SELECTED PROJECTS

Horuz V2 Modernization

149MKM Software · 2025 – Present · React, TypeScript, Vite, TanStack Virtual, Zustand, FastAPI, MongoDB, Docker

- Situation: V1 Horuz (CRA-based) faced performance, scalability, and UX consistency issues across six core modules. Action: **Led** end-to-end frontend modernization. Result: **~41%** total bundle reduction, **~67%** initial JS reduction, **~75%** faster loads, and stronger enterprise UX across Dashboard, Obligations, Frameworks, Auth, Actions, and Files.
- Accomplished safer, faster releases as measured by **~40%** deployment time reduction, by **improving CI/CD** and **Docker** workflows alongside modular V2 architecture.

Page-Segmented Virtualization System

149MKM Software / Horuz · 2025 – Present · React, TanStack Virtual, TypeScript, paginated FastAPI

- Situation: Compliance document tables exceeded standard virtual-list limits due to browser max scroll height and paginated APIs. Action: **Built** custom page-segmented virtualization combining **TanStack Virtual**, deduplication, cache-aware segments, and dynamic row sizing. Result: **200K+** records per **~20Mpx** segment; stress-tested near **1M** rows without freezing or data loss.

CRA-to-Vite Migration

149MKM Software / Horuz · 2025 · Vite, React, TypeScript, dynamic imports

- Accomplished **~6.6MB** → **~3.9MB** total bundle and **~3MB** → **~998KB** initial JS as measured by bundle analysis, by migrating from Create **React** App to **Vite** with route-level splitting, lazy loading, and caching strategy improvements.

Zustand Auth & Global State Refactor

149MKM Software / Horuz · 2025 · Zustand, React, TypeScript, JWT, RBAC

- Replaced Context-heavy auth with **Zustand** stores, structured bootstrapping, session lifecycle management, and role-aware routing — enabling scalable **JWT/RBAC** patterns and a clear path to future **MFA** without a full rewrite.

Files Module & Document Infrastructure

149MKM Software / Horuz · 2025 – Present · React, TanStack Virtual, AWS, FastAPI

- Accomplished **~50%** fewer redundant API calls and **~75%** faster document loads as measured by production monitoring, by implementing viewport virtualization, lazy metadata loading, **AWS** request caching, Excel exports, and secure downloads for enterprise compliance users.

Rights Administration (RA) Platform

149MKM Software / Horuz · 2025 · React, TypeScript, JWT, RBAC, Docker, CI/CD

- **Designed** a cross-service admin platform centralizing permissions and responsibilities across Horuz services with environment-aware **Docker** deployments — **improving** release safety and access governance for multi-tenant enterprise customers.

PCC — Protected Card Capture SDK

Paybilt / Allianz · Dec. 2024 · Vanilla JavaScript, CSS, payment APIs

- Situation: **\$1M+** Allianz partnership at risk without an embeddable payment capture SDK. Action: Single-handedly **delivered** **~5,000** lines of Vanilla JS in **~48 hours** with theming, prefill, field lock/hide, and API-triggered payments (card, e-Transfer, digital cheque). Result: Partnership preserved; non-technical teams launch embedded forms in under 10 minutes (**~70%** faster onboarding).

OCC V1 — Conservative Party of Canada Donations

Paybilt · Apr. 2024 · TypeScript, Next.js, React, REST

- Took over frontend ownership of the donation platform powering CPC fundraising — enabling **\$500K+** in **processed** contributions through secure, scalable checkout **integrated** with Paybilt payment infrastructure.

OCC V2 — One-Click Checkout

Paybilt · 2024 – 2025 · React, Next.js, TypeScript

- **Led** a 4-engineer frontend team to ship a PayPal-style one-click checkout product — **increasing** conversion rates by **25%** through **streamlined** payment UX and reusable checkout components.

iGamingOntario Name-Matching Compliance Tool

Paybilt · Nov. 2024 · Jinja2, Flask, payout APIs, MongoDB

- **Built** a **Jinja2**-powered compliance UI with **Flask** backend logic validating beneficiary names against registration data before e-Transfer payouts — intercepting mismatched transactions for regulated merchants (Bet365, BetMGM) with thousands of weekly payouts in production.

Transaction Tracking System

Paybilt · May 2024 · Jinja2, Flask, MongoDB

- Accomplished **~90%** less frontend template duplication as measured by template count across products, by **architecting** one adaptive **Jinja2** template for **300+** transaction state combinations — decreasing unclear-status support tickets by **~40%**.

Template Registry Service

Paybilt · 2024 – 2025 · FastAPI, MongoDB, Next.js, GraphQL, TypeScript, Jinja2

- **Engineered** an internal template management API and UI powering **100+** versioned **Jinja2** templates — enabling flexible, versioned rendering across Paybilt's product portfolio.

Zam Zam Bags and Boutique Website — [Live site](#)

Independent · Feb. 2023 – May 2023 · React.js, Firebase, Firestore, Bootstrap

- **Delivered** a responsive small-business e-commerce site with owner-managed catalogue CRUD, manual QA, and optimized Firebase operations — **improving** day-to-day business workflows without ongoing developer dependency.

Oasis Real Estate Platform

Oasis (Contract) · 2025 · Flutter, BLoC, WebSockets, REST, Google Maps

- **Shipped** a 22+ screen cross-platform real estate product from scratch with real-time chat, maps, and performance optimizations (**~60%** fewer API calls, **~70%** smaller images) for production release in the KSA market.

TECHNICAL HIGHLIGHTS

- Frontend architecture: reusable component libraries, route-level code splitting, lazy loading, separation of UI/state/side effects, and enterprise patterns with **Ant Design**, **Radix UI**, and **Shadcn/UI**.
- Performance: **CRA-to-Vite** migration, tree shaking, bundle analysis, intelligent API caching, request deduplication, and viewport-based virtualization for compliance document tables at enterprise scale.
- Auth & security: **Zustand**-based global state, structured auth bootstrapping, **JWT/RBAC**, role-aware routing, and multi-environment **Docker** deployments (development, staging, production).
- Payments & compliance: embeddable Vanilla JS SDKs, **Jinja2/Flask** compliance tooling, adaptive transaction-status rendering, and payout API integration for regulated merchants.
- Real-time systems: WebSocket dashboards for live monitoring at Horuz; WebSocket chat with typing indicators, read receipts, and reconnection in **Flutter** (Oasis).
- API integration: **REST** across **FastAPI/Flask**, **GraphQL** for internal template services, paginated loading paired with virtual tables, and secure **AWS**-backed file workflows.

SKILLS

- Frontend:** React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Next.js, Vite, Tailwind CSS, Bootstrap, Ant Design, Shadcn/UI, Radix UI, TanStack Virtual, Zustand, React Router, Chart.js, Recharts, i18next, Ajax, compound components, atomic design, WCAG
- Backend:** Python, FastAPI, Flask, Node.js, Express, REST APIs, GraphQL, WebSockets, Jinja2, JWT, RBAC, session management, auth bootstrapping
- Data & Cloud:** MongoDB, PostgreSQL, MySQL, Firestore, Firebase, AWS-hosted files, Google Cloud, document querying, paginated APIs
- Mobile:** Flutter, Dart, Clean Architecture, BLoC, Repository pattern, Google Maps, iOS, Android, cross-platform Web
- DevOps & Tools:** Docker, CI/CD, Git, GitHub, JIRA, Vercel, Nginx, environment-aware deployments, Figma, Adobe Photoshop, Linux, Bash
- Languages:** JavaScript, TypeScript, Python, Dart, C, C++, C#, .NET, SQL, HTML, CSS, UML
- Other:** Agile/SCRUM, TDD concepts, manual testing & debugging, Google Analytics, Microsoft Office, SharePoint, SAP, macOS, Windows

LEADERSHIP & COLLABORATION

- Promoted four times in twelve months at Paybilt by **shipping** production systems under executive and client pressure — from junior frontend intern to full stack developer owning SDK, compliance, and checkout roadmaps.
- **Led** a 4-engineer frontend squad for OCC V2 while serving as lead frontend architect at 149MKM — partnering with CTO, product owners, backend, QA, and design across distributed Agile delivery.
- Known for rapid learning and end-to-end feature ownership: translating business requirements into technical specs, **delivering** MVPs on 48-hour timelines, and supporting releases through documentation, code review, and post-deploy monitoring.

EDUCATION

Seneca Polytechnic <i>Bachelor of Technology – Software Development</i>	Apr. 2026 <i>Toronto, ON</i>
Certifications	
Responsive Web Design — freeCodeCamp	Sept. 2023
Front End Development Libraries — freeCodeCamp	May 2024
Backend Development and APIs — freeCodeCamp	Jan. 2025